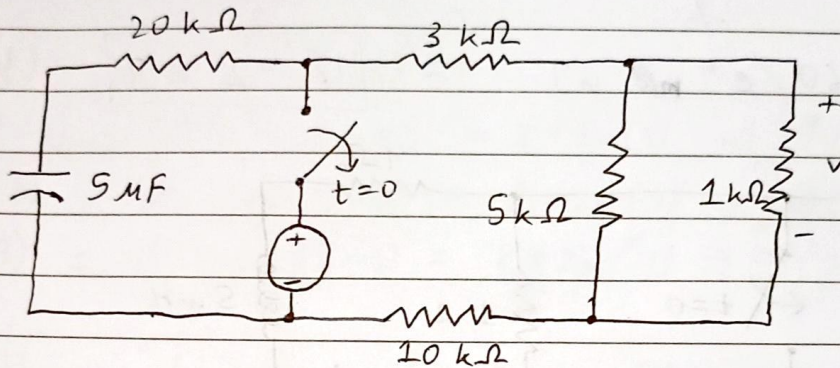


NESS M2023

HW2: Circuits

CHAPTER - 8 :

Q22.



$$\begin{aligned} a) R_{eq} &= \left(\frac{1}{5k\Omega} + \frac{1}{1k\Omega} \right)^{-1} + 20k\Omega + 3k\Omega + 10k\Omega \\ &= \left[\frac{5000}{6} + 20000 + 3000 + 10000 \right] k\Omega \\ &= \left[33 + \frac{5}{6} \right] k\Omega \approx 33.83333 k\Omega \end{aligned}$$

$$\begin{aligned} \therefore \text{Time constant, } \tau &= R_{eq} \cdot C = (33833.33 \times 5 \times 10^{-6}) s \\ &\Rightarrow \boxed{\tau = 169.15 \times 10^{-3} ms} \end{aligned}$$

- b) In such a long time, when wheels have been successfully employed into automation and locomotion, the capacitor must have charged sufficiently to practically break the ch.
- $\Rightarrow$ At $t = 0$, voltage across $1k\Omega$:-

$$\begin{aligned} v_t /_{t=0} &= \frac{(12V) \left[\frac{1}{5k\Omega} + \frac{1}{1k\Omega} \right]^{-1}}{R_{eq}} = \frac{12 \times 5000 \times 6}{203 \times 6 \times 1000} V \\ &= 0.2955 V \end{aligned}$$

$$\therefore \boxed{v_t = 0.2955 e^{-t/(169.15 \times 10^{-3})} V}$$

$\therefore$ For RC ckt's, voltage drops exponentially with time
And $V = V_0 e^{-t/\tau}$

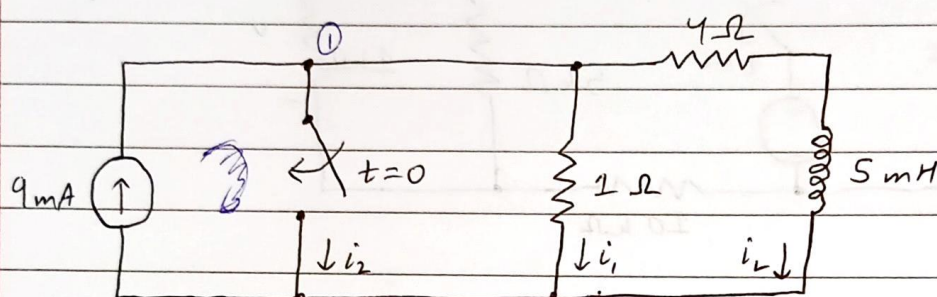
c) $V_t / \text{capacitor} = 12e^{-t/\tau}$, since $V_0 / \text{cap.} = 12 \text{ V}$.

$\Rightarrow$ Energy, $E_t = \frac{1}{2} \times C \times V_t^2$

$\Rightarrow E_{t=170 \text{ ms}} = \frac{1}{2} \times 5 \times 10^{-6} \times 144 \times e^{-2 \times 170 / 169.15}$

$= 360 / e^2 \text{ nJ} \approx 48.72 \text{ nJ}$

Q31.



a) At $t \geq 0$, both terminals of 1Ω are shorted $\Rightarrow i_1 = 0 \text{ A } \forall t$.

$\Rightarrow \boxed{i_1|_{t=1 \text{ ms}} = 0 \text{ A}}$

For a series LR ckt, $i_L|_t = I_0 e^{-Rt/L}$.

$I_0 = i_L|_{t=0} = \frac{(0.009 \text{ mA}) \times \left(\frac{1}{4} + \frac{1}{1} \right)^{-1}}{4 \Omega} = 1.8 \text{ mA}$

$\therefore i_L|_{t=1 \text{ ms}} = 1.8 e^{-4t/0.005} \text{ mA}$

From KCL at node 1: $i_2|_t = (9 - i_L|_t) \text{ mA}$

$\Rightarrow i_2|_{t=1 \text{ ms}} = 9 - 1.8 e^{-4/5} \text{ mA} \Rightarrow \boxed{i_2|_{t=1 \text{ ms}} = 8.191 \text{ mA}}$

b) $\boxed{i_1|_{t=3 \text{ ms}} = 0 \text{ A}}$

$i_2|_{t=3 \text{ ms}} = 9 - 1.8 e^{-12/5} \text{ mA}$

$\Rightarrow \boxed{i_2|_{t=3 \text{ ms}} = 8.837 \text{ mA}}$

Q42. a) $f(t) = t u(1-t)$

$$[u(t) = 0 \quad \forall t < 0, 1 \quad \forall t \geq 0]$$

$$\Rightarrow f(-1) = -1 - u(1+1) \Rightarrow \boxed{f(-1) = -1}$$

$$\Rightarrow f(0) = 0 u(1) \Rightarrow \boxed{f(0) = 0}$$

$$\Rightarrow f(3) = 3 u(1-3) = 3 u(-2) \Rightarrow \boxed{f(3) = 0}$$

b) $g(t) = 8 + 2u(2-t)$

$$\Rightarrow g(-1) = 8 + 2u(2+1) = 8 + 2u(3) \Rightarrow \boxed{g(-1) = 10}$$

$$\Rightarrow g(0) = 8 + 2u(2-0) = 8 + 2u(2) \Rightarrow \boxed{g(0) = 10}$$

$$\Rightarrow g(3) = 8 + 2u(2-3) = 8 + 2u(-1) \Rightarrow \boxed{g(3) = 8}$$

c) $h(t) = u(t+1) - u(t-1) + u(t+2) - u(t-4)$

$$\Rightarrow h(-1) = u(-1+1) - u(-1-1) + u(-1+2) - u(-1-4)$$

$$= u(0) - u(-2) + u(1) - u(-5) \Rightarrow \boxed{h(-1) = 2}$$

$$\Rightarrow h(0) = u(0+1) - u(0-1) + u(0+2) - u(0-4)$$

$$= u(1) - u(-1) + u(2) - u(-4) \Rightarrow \boxed{h(0) = 2}$$

$$\Rightarrow h(3) = u(3+1) - u(3-1) + u(3+2) - u(3-4)$$

$$= u(4) - u(2) + u(5) - u(-1) \Rightarrow \boxed{h(3) = 1}$$

d) $z(t) = 1 + u(3-t) + u(t-2)$

$$\Rightarrow z(-1) = 1 + u(3+1) + u(-1-2)$$

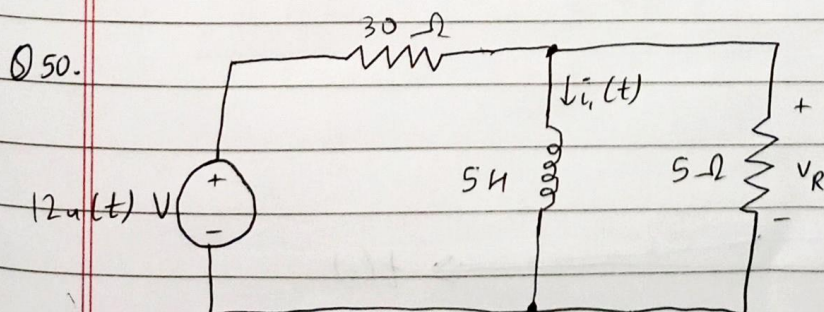
$$= 1 + u(4) + u(-3) \Rightarrow \boxed{z(-1) = 2}$$

$$\Rightarrow z(0) = 1 + u(3-0) + u(0-2)$$

$$= 1 + u(3) + u(-2) \Rightarrow \boxed{z(0) = 2}$$

$$\Rightarrow z(3) = 1 + u(3-3) + u(3-2)$$

$$= 1 + u(0) + u(1) \Rightarrow \boxed{z(3) = 3}$$



$$a) R_{eq} = \left(\frac{1}{30} + \frac{1}{5} \right)^{-1} = \frac{30}{7} \Omega$$

$$\Rightarrow \tau = \frac{L}{R_{eq}} = \frac{5 \times 2}{30} = \frac{2}{6} s = 1.167 s$$

(Time constant)

$$\text{Now, } i(t) = Ae^{-t/\tau} + i_L(\infty) \quad [i_n(t) + i_s(t)]$$

$$\text{And, } i_L(\infty) = \frac{12}{30} A = 0.4 A \quad (\text{inductor short-circuited})$$

$$\text{Also, } V(t) = 12u(t) = 0 \quad \forall t < 0.$$

$$\Rightarrow i(t) = 0 \quad \forall t < 0 \Rightarrow i_L(\infty) + Ae^{-t/\tau} = 0 \quad \forall t < 0$$

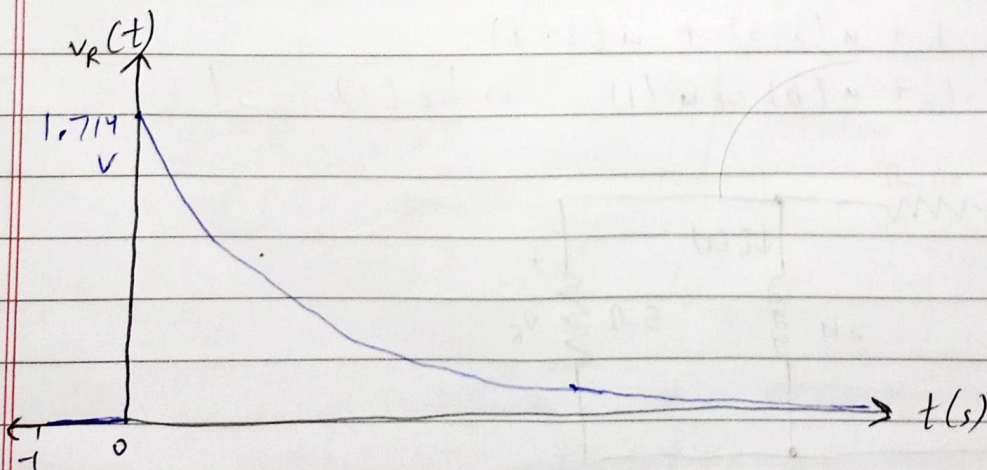
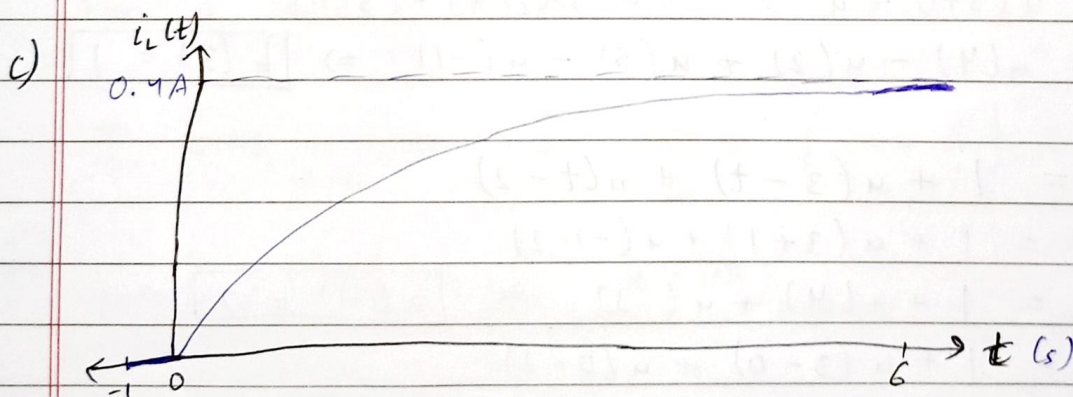
$$\Rightarrow A = -i_L(\infty) = -0.4 A.$$

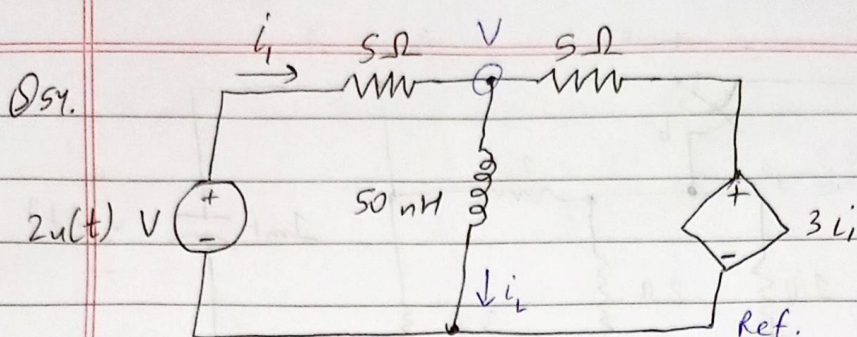
$$\therefore i_L(t) = 0.4 - 0.4e^{-t/1.167} A$$

$$b) v_R(t) = v_L(t) = L \frac{d}{dt} [i(t)]$$

$$= 5 \cdot \frac{d}{dt} [0.4 - 0.4e^{-t/1.167}] = \frac{5 \times 0.4}{1.167} e^{-t/1.167}$$

$$\therefore v_R(t) = 1.714 e^{-t/1.167} V$$





For $t \leq 0$, both voltage source and inductor act as short ckt.

Also, $i_L(0) = 0$.

For $t = 0$: $2 - 10i_L(0) - 3i_L(0) = 0$ [KVL in supermesh]

$$\Rightarrow i_L(0) = \frac{2}{13} \text{ A} \quad \text{--- (1)}$$

Using KCL at given node, replacing inductor with test current:

$$\frac{V}{5} - 1 + \frac{V + \frac{3V}{5}}{5} = 0 \quad (1 \text{ A} = i_L)$$

$$\Rightarrow V = 1.9231 \text{ V}$$

Also, resistance seen by inductor $R_L = \frac{V}{i_L} = V = 1.9231 \Omega$

$$\therefore \text{Time constant, } \tau = \frac{L}{R_L} = \frac{50 \times 10^{-9}}{1.9231} = 26 \text{ ns.}$$

$$\text{As } t \rightarrow \infty, i_L(t) = 0 \Rightarrow \boxed{i_L(\infty) = 0.4 \text{ A}} \quad (\text{using KCL})$$

Final expression for $i_L(t) = i_{in}(t) + i_L(\infty)$

where, $i_{in}(t) = k e^{-t/\tau}$

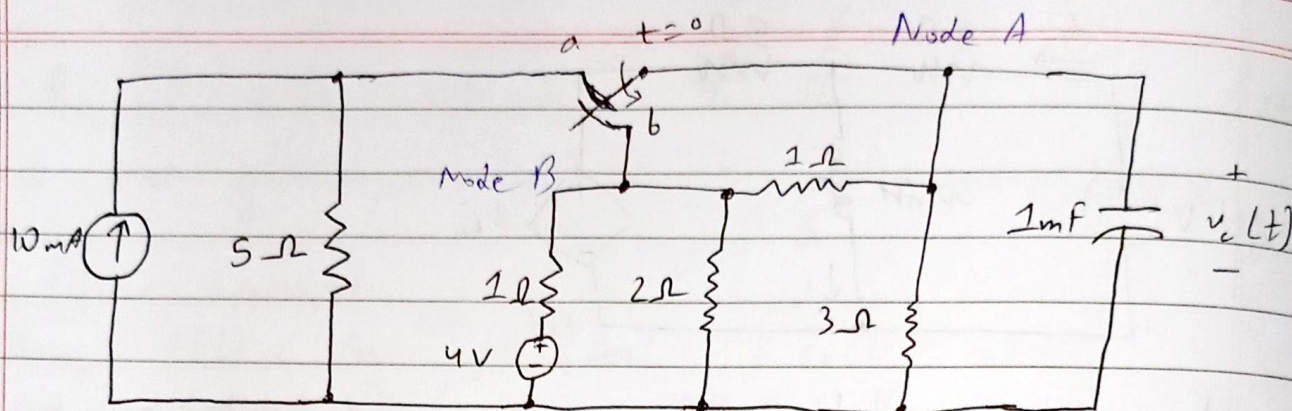
However, $i_L(0) = \frac{2}{13} \text{ A}$ (from (1))

$$\Rightarrow i_L(0) = k e^{-t/\tau} i_L(\infty) + k = \frac{2}{13}$$

$$\Rightarrow k = \frac{2}{13} - 0.4 = -0.246$$

$$\therefore \boxed{i_L(t) = -0.246 e^{\frac{-t}{(26 \times 10^{-9})}} + 0.4 \text{ A}}$$

Q 61.



$t < 0$: The capacitor is open-circuited.

KCL at Node A:

$$\frac{V_A}{5} + \frac{V_A}{3} + \frac{V_A - V_B}{1} - 10 \times 10^{-3} = 0 \quad \text{--- (1)}$$

KCL at Node B:

$$\frac{V_B - V_A}{1} + \frac{V_B}{2} + \frac{V_B - 4}{1} = 0$$

$$\Rightarrow 2V_B - 2V_A + V_B + 2V_B - 8 = 0$$

$$\Rightarrow 5V_B - 2V_A = 8 \quad \Rightarrow V_B = \frac{8 + 2V_A}{5}$$

Using this in eq. (1):

$$\frac{V_A}{5} + \frac{V_A}{3} + V_A - \frac{8 + 2V_A}{5} = 10 \times 10^{-3}$$

$$\Rightarrow 3V_A + 5V_A + 15V_A - 24 - 6V_A = 150 \times 10^{-3}$$

$$\Rightarrow 17V_A = 24.15 \quad \Rightarrow V_A = 1.42 \text{ V}$$

Now, $v_c(t) = V_A = 1.42 \text{ V} \quad \forall t < 0$.

$$P_{5\Omega}|_{t=0^-} = \frac{V_A^2}{5} = \frac{1.42 \times 1.42}{5} = 0.4 \text{ W}$$

$t \rightarrow \infty$: Capacitor is open-circuited. Also $V_A = V_B$.

KCL at Node B:

$$\frac{V_B - 4}{1} + \frac{V_B}{2} + \frac{V_B}{3} = 0$$

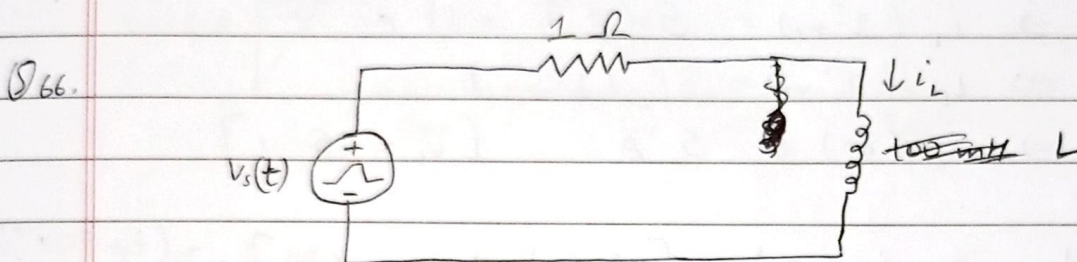
$$\Rightarrow 6V_B - 24 + 3V_B + 2V_B = 0 \Rightarrow V_B = \frac{24}{11} = 2.18 \text{ V}$$

$$v_c(\infty) = V_A = V_B = 2.18 \text{ V} = v_c(\infty)$$

Time constant, $\tau = R_{eq} \cdot C$
 $= \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} \right)^{-1} \times 10^{-3} \text{ s} = 0.545 \text{ ms}$

For $t \geq 0$, $v_c(t) = v_c(\infty) + [v_c(0^+) - v_c(\infty)] e^{-t/\tau}$
 $= 2.18 + (1.42 - 2.18) e^{-t/\tau}$
 $\Rightarrow v_c(t) = 2.18 - 0.76 e^{-1834.86 t} \text{ V}$

$\therefore P_{3\Omega} \big|_{t=2\text{ms}} = \frac{[v_c(2\text{ms})]^2}{3\Omega} = \boxed{1.56 \text{ W}}$



~~$L = 100 \text{ mH}$~~

a) $v_s(t) = 5u(t) - 5u(t - 10^{-9}) + 5u(t - 2 \times 10^{-9}) \text{ V}$;
 $0 \leq t \leq 4 \text{ ns}$

$\Rightarrow v_s(t) = \begin{cases} 0, & t < 0 \\ 5, & 0 \leq t \leq 1 \text{ ns} \\ 0, & 1 \leq t \leq 2 \text{ ns} \\ 5, & 2 \leq t \leq 4 \text{ ns} \end{cases} \text{ V}$

$\tau = \frac{L}{R} = 1 \text{ ns} \quad [R_{eq} = 1 \Omega]$

$t = 0^- \Rightarrow i_L(0^-) = 0 \text{ A}$

$t = 0^+ \Rightarrow i_L(0^+) = 0 \text{ A} = i_L(0^-)$

$t \rightarrow \infty \Rightarrow i_L(\infty) = 5 \text{ A (short-ckt.)} \quad [v_s = 5 \text{ V}]$

$\therefore i_L(t) = i_L(\infty) + [i_L(0^+) - i_L(\infty)] e^{-t/\tau}$
 $[0 \leq t \leq 1 \text{ ns}]$
 $= 5 - 5e^{-t/1} \text{ A}$

-①

$$t = 1^- \text{ ns} \Rightarrow i_L(1^- \text{ ns}) = 5 - 5e^{-10^{-9}/L}$$

$$t = 1^+ \text{ ns} \Rightarrow i_L(1^+ \text{ ns}) = i_L(1^- \text{ ns})$$

$$t \rightarrow \infty \Rightarrow i_L(\infty) = 0 \text{ A} \quad [V_s = 0 \text{ V}]$$

$$\therefore i_L(t) = i_L(\infty) + [i_L(1^+ \text{ ns}) - i_L(\infty)]e^{-(t-10^{-9})/L}$$

$$[1 \text{ ns} \leq t \leq 2 \text{ ns}]$$

$$= 5(1 - e^{-10^{-9}/L})e^{-(t-10^{-9})/L} = 5(1 - e^{-\frac{10^{-9}}{L}})e^{-\frac{t}{L}} \cdot e^{\frac{10^{-9}}{L}}$$

$$= 5(e^{\frac{10^{-9}}{L}} - 1)e^{-t/L} \quad - (2)$$

$$t = 2^- \text{ ns} \Rightarrow i_L(2^- \text{ ns}) = 5(e^{\frac{10^{-9}}{L}} - 1)e^{-\frac{2 \times 10^{-9}}{L}}$$

$$t = 2^+ \text{ ns} \Rightarrow i_L(2^+ \text{ ns}) = i_L(2^- \text{ ns})$$

$$t \rightarrow \infty \Rightarrow i_L(\infty) = 5 \text{ A} \quad [V_s = 5 \text{ V}]$$

$$\therefore i_L(t) = i_L(\infty) + [i_L(2^+ \text{ ns}) - i_L(\infty)]e^{-(t-2 \times 10^{-9})/L}$$

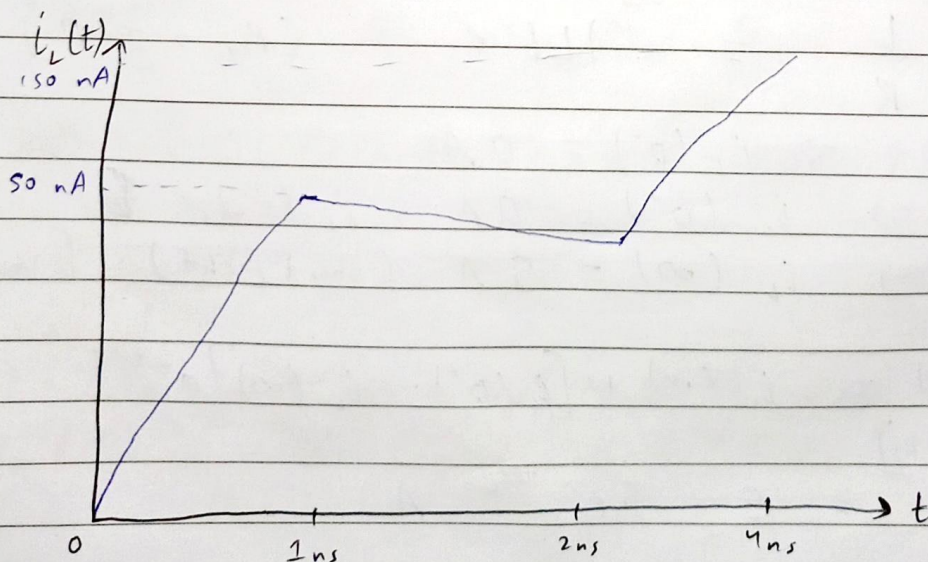
$$[2 \text{ ns} \leq t \leq 4 \text{ ns}]$$

$$= 5 + 5[e^{\frac{10^{-9}}{L}} - 1 - e^{\frac{2 \times 10^{-9}}{L}}]e^{-\frac{t}{L}} \cdot e^{\frac{2 \times 10^{-9}}{L}}$$

$$= 5 + 5(e^{\frac{10^{-9}}{L}} - 1 - e^{\frac{2 \times 10^{-9}}{L}})e^{-t/L} \quad - (3)$$

i) $L = 100 \text{ mH} \quad (= 10^{-1} \text{ H})$

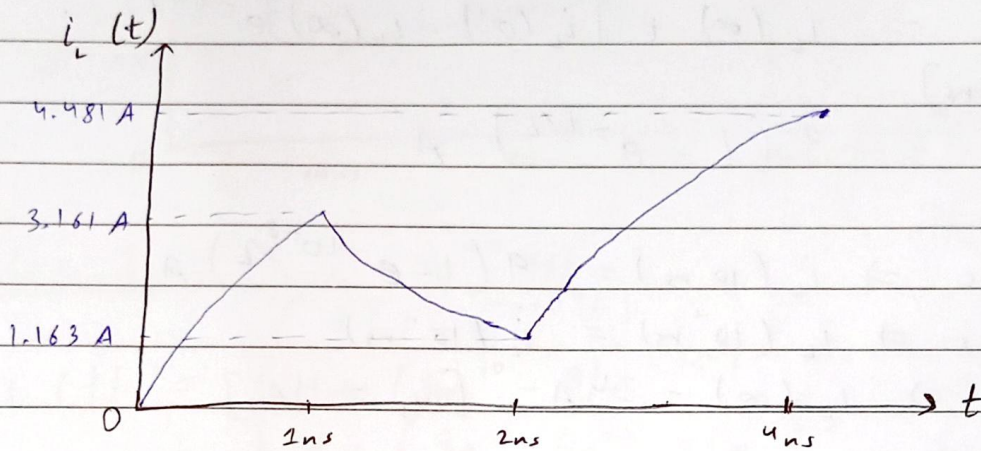
$$i_L(t) = \begin{cases} 5(1 - e^{-10^8 t}) & 0 \leq t \leq 1 \text{ ns} \\ 5(e^{10^{-8}} - 1)e^{-10^8 t} & 1 \text{ ns} \leq t \leq 2 \text{ ns} \\ 5 + 5(e^{10^{-8}} - e^{2 \times 10^{-8}} - 1)e^{-10^8 t} & 2 \text{ ns} \leq t \leq 4 \text{ ns} \end{cases} \quad A$$



ii) $L = 1 \text{ nH}$

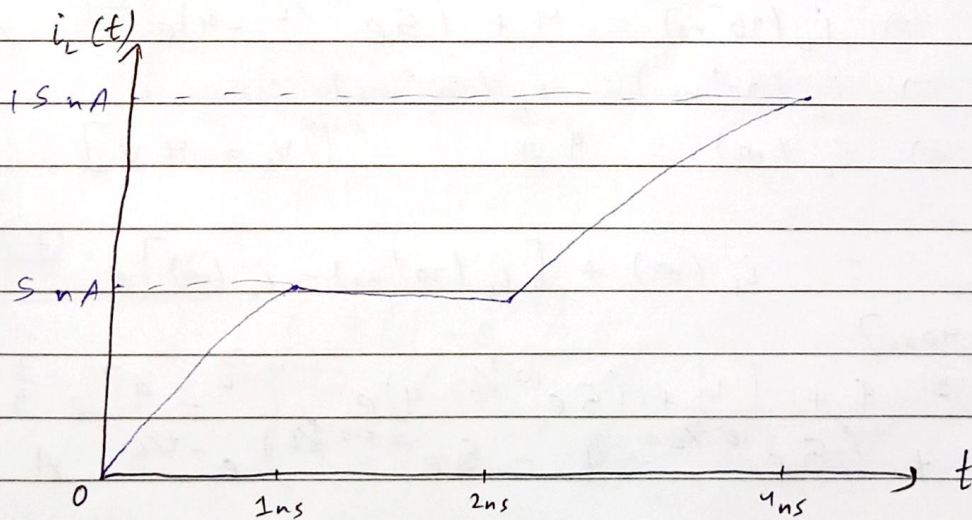
$(= 10^{-9} \text{ H})$

$$i_L(t) = \begin{cases} 5(1 - e^{-10^9 t}) & , 0 \leq t \leq 1 \text{ ns} \\ 5(e - 1)e^{-10^9 t} & , 1 \leq t \leq 2 \text{ ns} \\ 5 + 5(e - e^2 - 1)e^{-10^9 t} & , 2 \leq t \leq 4 \text{ ns} \end{cases} \quad A$$



iii) $L = 1 \text{ H}$

$$i_L(t) = \begin{cases} 5(1 - e^{-t}) & , 0 \leq t \leq 1 \text{ ms} \\ 5(e^{10^{-9}} - 1)e^{-t} & , 1 \leq t \leq 2 \text{ ms} \\ 5 + 5(e^{10^{-9}} - e^{2 \times 10^{-9}} - 1)e^{-t} & , 2 \leq t \leq 4 \text{ ms} \end{cases} \quad A$$



b) $v_s(t) = 9u(t) - 5u(t - 10^{-8}) + 5u(t - 2 \times 10^{-8}) \text{ V} \quad 0 \leq t \leq 40 \text{ ns}$

$$v_s(t) = \begin{cases} 9 \text{ V} & , 0 \leq t \leq 10 \text{ ns} \\ 4 \text{ V} & , 10 \text{ ns} \leq t \leq 20 \text{ ns} \\ 9 \text{ V} & , 20 \leq t \leq 40 \text{ ns} \end{cases}$$

$\tau = \frac{L}{R} ; R = 1 \Omega \Rightarrow \tau = 1 \text{ H s}$

$$t = 0^- \Rightarrow i_L(0^-) = 0A$$

$$t = 0^+ \Rightarrow i_L(0^+) = i_L(0^-) = 0A$$

$$t \rightarrow \infty \Rightarrow i_L(\infty) = 9A \quad [V_s = 9V; R = 1\Omega; \text{short-circuit}]$$

$$\begin{aligned} \therefore i_L(t) &= i_L(\infty) + [i_L(0^+) - i_L(\infty)]e^{-t/\tau} \\ [0 \leq t \leq 10\mu s] \\ &= 9(1 - e^{-t/L})A \end{aligned} \quad (1)$$

$$t = 10^- \mu s \Rightarrow i_L(10^- \mu s) = 9(1 - e^{-10^{-8}/L})A$$

$$t = 10^+ \mu s \Rightarrow i_L(10^+ \mu s) = i_L(10^- \mu s)$$

$$t \rightarrow \infty \Rightarrow i_L(\infty) = 4A \quad [V_s = 4A]$$

$$\begin{aligned} \therefore i_L(t) &= i_L(\infty) + [i_L(10^+ \mu s) - i_L(\infty)]e^{-(t - 10^{-8})/\tau} \\ [10\mu s \leq t \leq 20\mu s] \\ &= 4 + (9 - 9e^{-\frac{10^{-8}}{L}} - 4)e^{-\frac{t}{L}} \cdot e^{\frac{10^{-8}}{L}} \\ &= 4 + (5e^{10^{-8}/L} - 9)e^{-t/L}A \end{aligned} \quad (2)$$

$$t = 20^- \mu s \Rightarrow i_L(20^- \mu s) = 4 + (5e^{10^{-8}/L} - 9)e^{-\frac{2 \times 10^{-8}}{L}}A$$

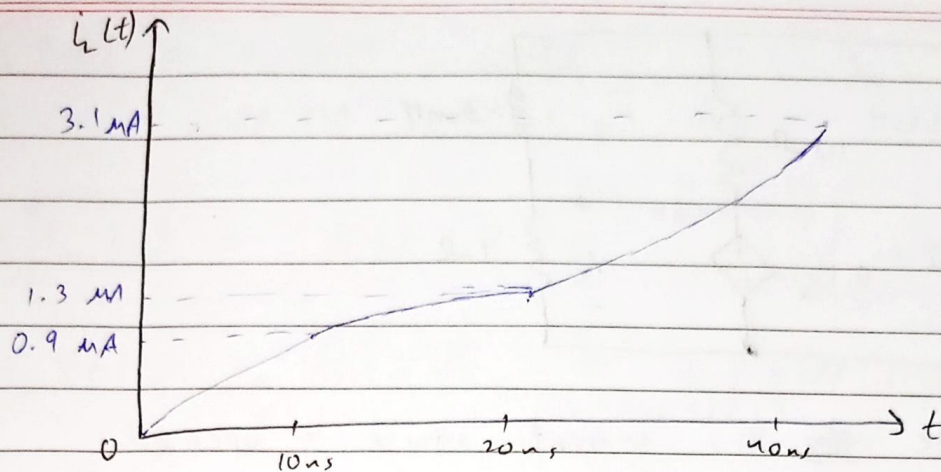
$$t = 20^+ \mu s \Rightarrow i_L(20^+ \mu s) = i_L(20^- \mu s)$$

$$t \rightarrow \infty \Rightarrow i_L(\infty) = 9A \quad [V_s = 9V]$$

$$\begin{aligned} \therefore i_L(t) &= i_L(\infty) + [i_L(20^+ \mu s) - i_L(\infty)]e^{-\frac{(t - 2 \times 10^{-8})}{\tau}} \\ [20\mu s \leq t \leq 40\mu s] \\ &= 9 + [4 + (5e^{10^{-8}/L} - 9)e^{-\frac{2 \times 10^{-8}}{L}} - 9]e^{-\frac{t}{L}} \cdot e^{\frac{2 \times 10^{-8}}{L}} \\ &= 9 + (5e^{10^{-8}/L} - 9 - 5e^{\frac{2 \times 10^{-8}}{L}})e^{-t/L}A \end{aligned} \quad (3)$$

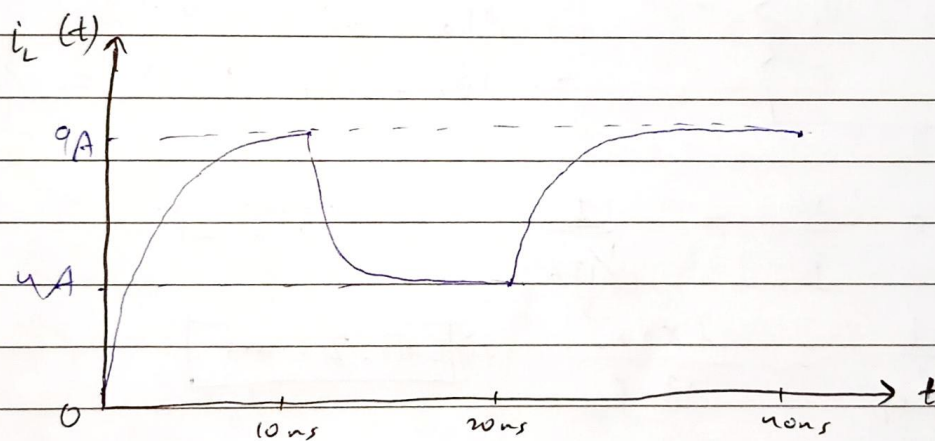
i) $L = 100 \text{ mH} \quad (= 10^{-1} \text{ H})$

$$i_L(t) = \begin{cases} 9(1 - e^{-10t}) & , 0 \leq t \leq 10\mu s \\ 4 + (5e^{10^{-7}} - 9)e^{-10t} & , 10 \leq t \leq 20\mu s \\ 9 + (5e^{10^{-7}} - 5e^{2 \times 10^{-7}} - 9)e^{-10t} & , 20 \leq t \leq 40\mu s \end{cases} A$$



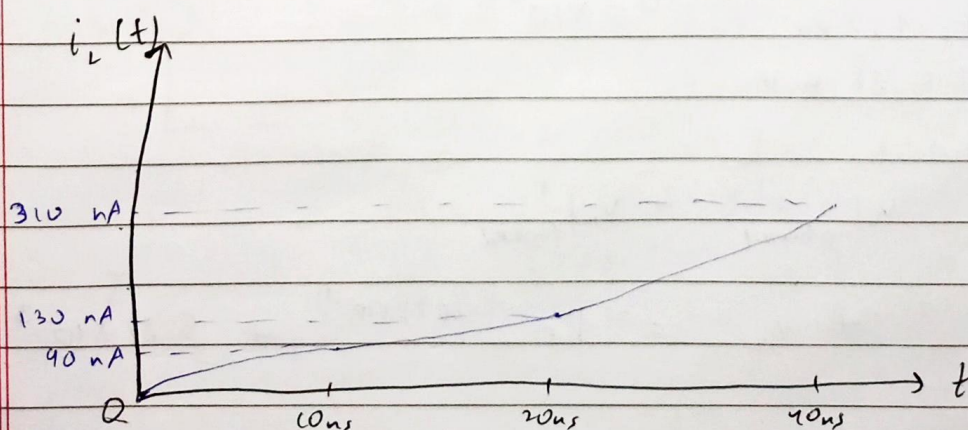
ii) $L = 1 \text{ nH}$ ($= 10^{-9} \text{ H}$)

$$i_L(t) = \begin{cases} 9(1 - e^{-10^9 t}) & , 0 \leq t \leq 10 \text{ ns} \\ 4 + (5e^{10} - 9)e^{-10^9 t} & , 10 \leq t \leq 20 \text{ ns} \\ 9 + (5e^{10} - 5e^{20} - 9)e^{-10^9 t} & , 20 \leq t, 40 \text{ ns} \end{cases} \quad A$$

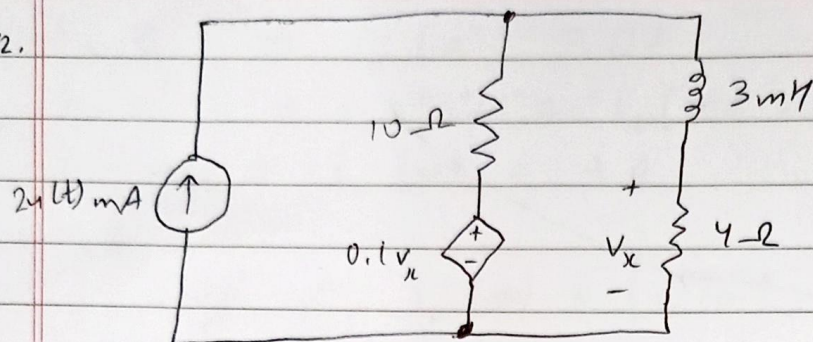


iii) $L = 1 \text{ H}$

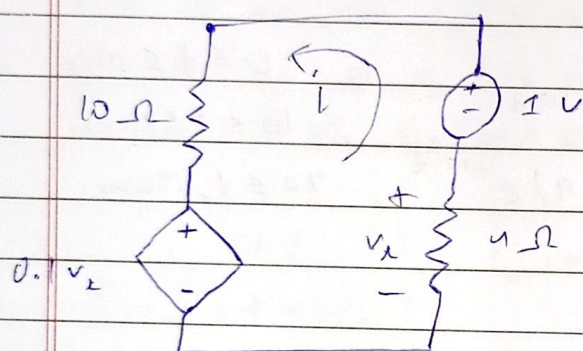
$$i_L(t) = \begin{cases} 9(1 - e^{-t}) & , 0 \leq t \leq 10 \text{ ns} \\ 4 + (5e^{10^{-8}} - 9)e^{-t} & , 10 \leq t \leq 20 \text{ ns} \\ 9 + (5e^{10^{-8}} - 5e^{2 \times 10^{-8}} - 9)e^{-t} & , 20 \leq t \leq 40 \text{ ns} \end{cases} \quad A$$



Q72.



- a) To compute R_{eq} , current source is opened.
To compute R_{eq} , inductor is replaced with $1V$ source.



KVL:

$$1 - 10i - 0.1V_x + V_x = 0$$

$$\Rightarrow 10i - 0.9V_x = 1$$

$$\text{Also, } V_x = -4i$$

$$\Rightarrow 10i + 3.6i = 1 \Rightarrow i = 1/13.6 \text{ A}$$

$$\therefore R_{eq} = \frac{V}{I} = \frac{1}{1/13.6} = 13.6 \Omega$$

$$\therefore \tau = \frac{L}{R_{eq}} = \frac{3 \times 10^{-3}}{13.6} \Rightarrow \tau = 0.22 \text{ ms}$$

b) $V_x = 0 \quad \forall t < 0$

As $t \rightarrow \infty$, inductor is short-circuited.

Using KCL, with bottom node as ref,

$$\frac{V_x}{4} + \frac{V_x - 0.1V_x}{10} = 2 \times 10^{-3}$$

$$\Rightarrow 5V_x + 1.8V_x = 40 \times 10^{-3}$$

$$\Rightarrow V_x = 5.88 \text{ mV}$$

$$\therefore V_x(t) = V_x|_{\text{natural}} + V_x|_{\text{forced}}$$

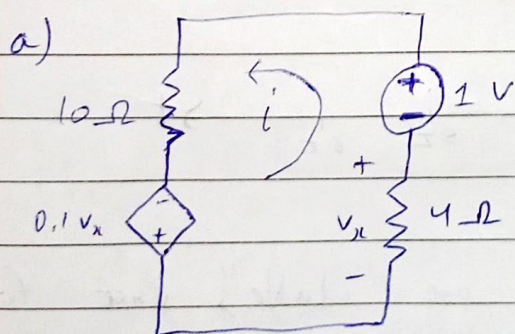
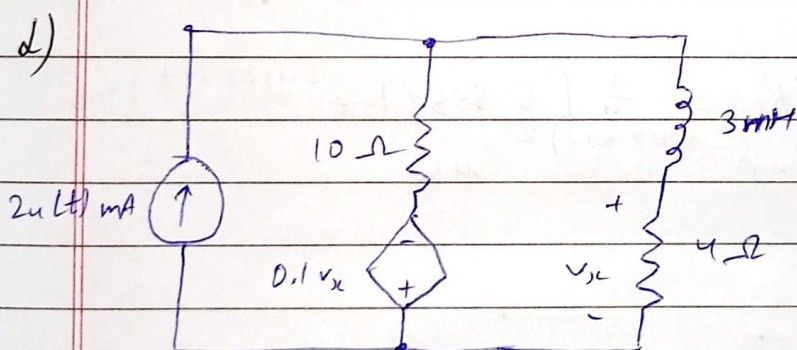
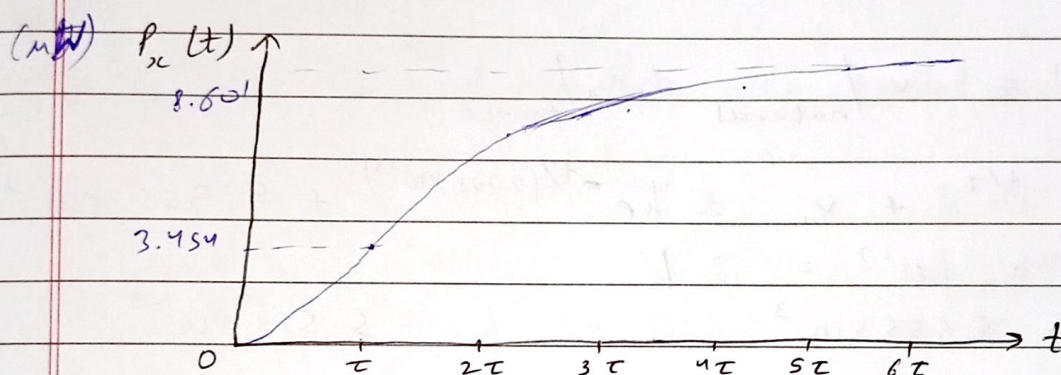
$$= ke^{-t/\tau} + V_x = ke^{-t/(0.22 \times 10^{-3})} + 5.88 \times 10^{-3}$$

$$v_x(0) = v_x(0^-) = 0 \text{ V}$$

$$\Rightarrow k + 5.88 \times 10^{-3} = 0 \Rightarrow k = -5.88 \times 10^{-3}$$

$$\Rightarrow v_x(t) = 5.88 (1 - e^{-4545.45t}) \text{ mV}$$

$$\begin{aligned} d) P_x(t) &= v_x^2(t) / R = \frac{1}{4} [5.88 (1 - e^{-4545.45t}) \times 10^{-3}]^2 \text{ W} \\ &= 8.6436 (1 - e^{-4545.45t})^2 \text{ mW} \end{aligned}$$



KVL:

$$1 - 10i + 0.1v_x + v_x = 0$$

$$\Rightarrow 10i - 1.1v_x = 1$$

$$\text{Also, } v_x = -4i$$

$$\Rightarrow 10i + 4.4i = 1 \Rightarrow i = 1/14.4 \text{ A}$$

$$\therefore R_{eq} = \frac{V}{I} = \frac{1}{1/14.4} = 14.4 \Omega$$

$$\therefore \tau = \frac{L}{R_{eq}} = \frac{3 \times 10^{-3}}{14.4} \Rightarrow \tau = 0.208 \text{ ms}$$

b) $v_x = 0 \quad \forall t < 0$

As $t \rightarrow \infty$, inductor is short-circuited.

Using KCL, (ref. at bottom node);

$$\frac{v_x + 0.1v_x}{10} + \frac{v_x}{4} = 2 \times 10^{-3}$$

$$\Rightarrow 2.2v_x + 5v_x = 40 \times 10^{-3}$$

$$\Rightarrow v_x = 5.555 \text{ mV}$$

$$\therefore v_x(t) = v_{x/\text{natural}} + v_{x/\text{forced}}$$

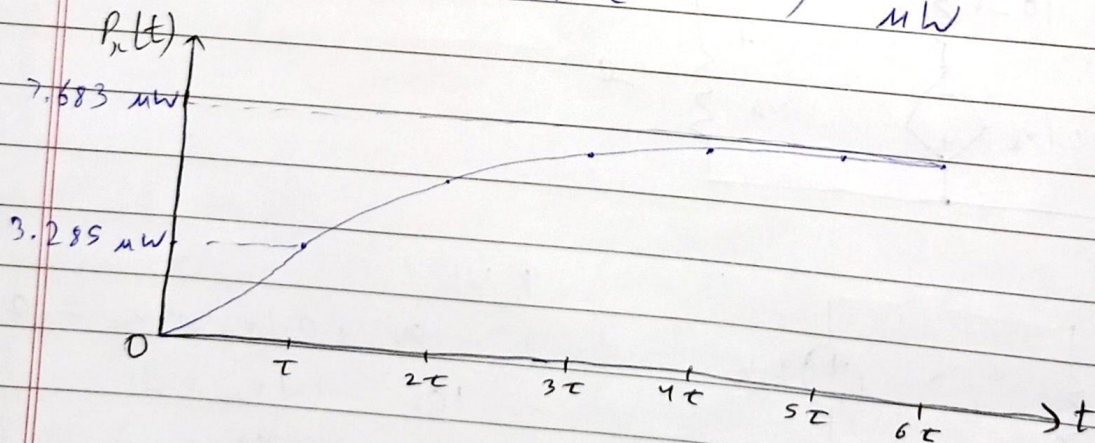
$$= ke^{-t/\tau} + v_x = ke^{-t/(0.208 \times 10^{-3})} + 5.555 \times 10^{-3}$$

$$v_x(0) = v_x(0^-) = 0 \text{ V}$$

$$\Rightarrow k + 5.555 \times 10^{-3} = 0 \Rightarrow k = -5.555 \times 10^{-3}$$

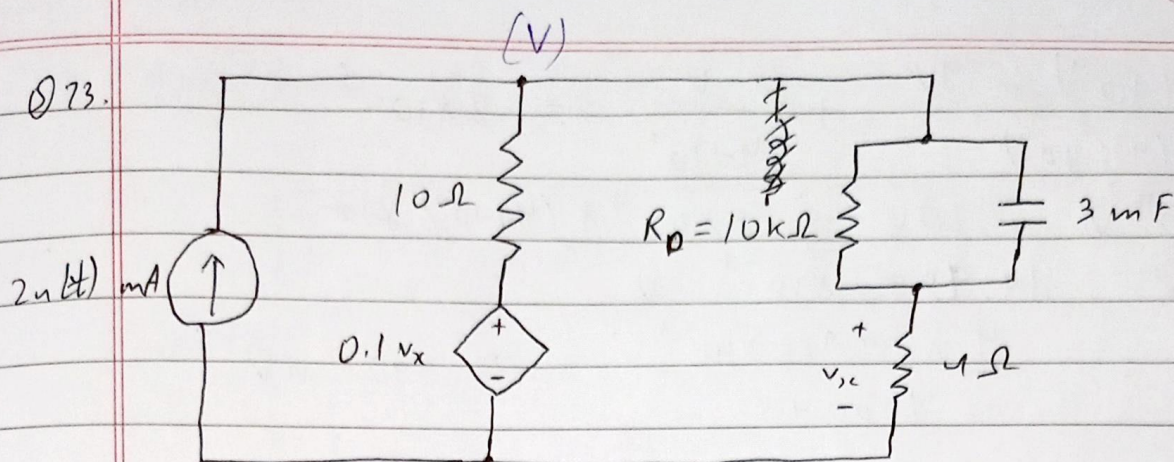
$$\Rightarrow v_x(t) = 5.555(1 - e^{-4807.69t}) \text{ mV}$$

$$\begin{aligned} \text{c) } P_x(t) &= v_x^2(t) / R = \frac{1}{4} [5.555(1 - e^{-4807.69t}) \times 10^{-3}]^2 \text{ W} \\ &= 7.71(1 - e^{-4807.69t})^2 \text{ mW} \end{aligned}$$



e) Yes, both circuit configurations are "stable", since for bounded inputs, the output does not blow.

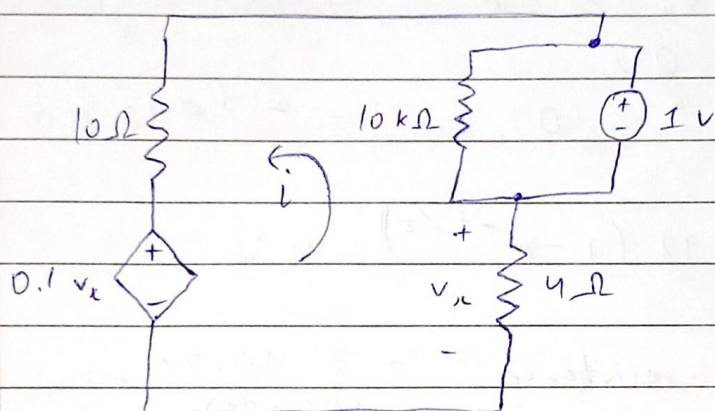
Q73.



a) $R_p = 10 \text{ k}\Omega$

3 mF with

To calculate R_{eq} , open current src. and replace 1 V .



KVL:

$$1 - 10i - 0.1v_x + v_x = 0$$

$$\Rightarrow 10i - 0.9v_x = 1$$

$$\text{Also, } v_x = -4i$$

$$\Rightarrow 10i + 3.6i = 1$$

$$\Rightarrow i = \frac{1}{13.6} \text{ A}$$

$$\Rightarrow R_1 = \frac{V}{i} = 13.6 \Omega$$

$$\therefore R_{eq} = \left(\frac{1}{13.6} + \frac{1}{10000} \right)^{-1} = 13.58 \Omega$$

$$\therefore \tau = R_{eq}C = 13.58 \times 3 \times 10^{-3} \Rightarrow \tau_1 = 40.74 \text{ ms}$$

$$\bullet R_p = \infty \Rightarrow R_{eq} = R_1 = 13.6 \Omega$$

$$\Rightarrow \tau_2 = 13.6 \times 3 \times 10^{-3} \Rightarrow \tau_2 = 40.8 \text{ ms}$$

$$\Delta = |\tau_1 - \tau_2| = 0.06 \text{ ms} = 60 \mu\text{s}$$

b) $v_x = 0 \quad \forall \quad t < 0$

As $t \rightarrow \infty$, capacitor is open-circuited.

Using KCL with ref. at bottom node,

$$\frac{V - 0.1v_x}{10} + \frac{V}{4 + 10^4} = 2 \times 10^{-3}$$

$$\text{Also, } v_x = \frac{4V}{4 + 10^4} \quad (\text{voltage division})$$

$$\frac{V(4+10^4) - 4V}{10(4+10^4)} + \frac{V}{4+10^4} = 2 \times 10^{-3}$$

$$\Rightarrow 10^4 V + 10V = 2 \times 10^{-3} \times (4+10^4) \times 10$$

$$\Rightarrow V = 19.988 \times 10^{-3} \text{ V}$$

$$\therefore V_{\infty} = \frac{4 \times 19.988 \times 10^{-3}}{10004} = 7.992 \text{ mV}$$

$$\therefore v_x(t) = v_x|_{\text{natural}} + v_x|_{\text{forced}}$$

$$= ke^{-t/\tau} + v_{\infty} = ke^{-t/\tau} + 7.992 \times 10^{-6}$$

$$v_x(0) = v_x(0^-) = 0 \text{ V}$$

$$\Rightarrow k + 7.992 \times 10^{-6} = 0 \Rightarrow k = -7.992 \times 10^{-6}$$

$$\Rightarrow v_x(t) = 7.992 (1 - e^{-t/\tau}) \text{ mV}$$

• Considering dielectric resistance, $\tau = 40.74 \text{ ms}$,

$$\Rightarrow v_x(200 \text{ ms}) = 7.992 (1 - e^{-200/40.74}) \text{ mV}$$

$$= 7.933 \text{ mV}$$

• With $R_p = \infty$, $\tau = 40.8 \text{ ms}$

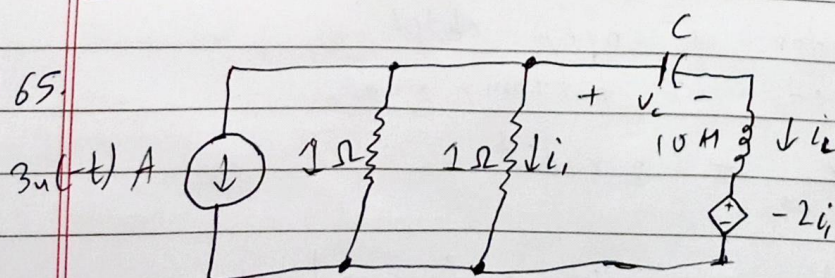
$$\Rightarrow v_x(200 \text{ ms}) = 7.992 (1 - e^{-200/40.8}) \text{ mV}$$

$$= 7.932 \text{ mV}$$

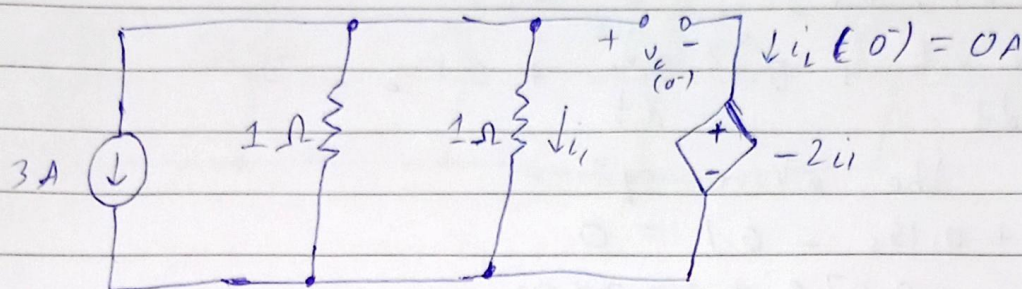
The dielectric resistance ^{merely} does not affects the resistance voltage, with a meagre difference of 1 mV.

CHAPTER-9

Q 65.



For $t < 0$, At $t = 0^-$, the circuit is in a steady state.



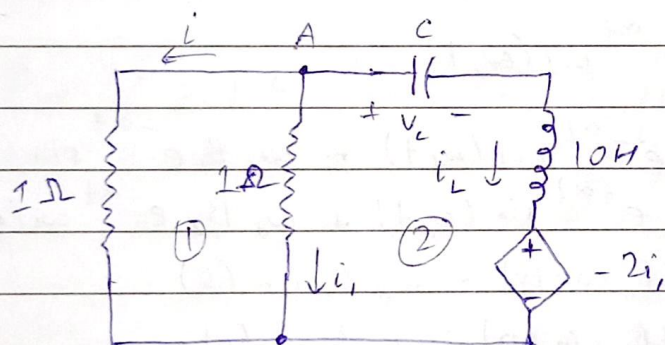
$$i_1 = \frac{-3 \times 1}{1+1} \text{ A} = -1.5 \text{ A} \quad (\text{current division})$$

KVL in ~~left~~ rightmost loop :-

$$v_c(0^-) - i_1 - 2i_1 = 0$$

$$\Rightarrow v_c(0^-) = 3i_1 = -4.5 \text{ V}$$

At $t > 0$, the circuit is as below:



KVL in loop 2 :-

$$-2i_1 + 10 \frac{di_L}{dt} + v_c - i_1 = 0$$

$$\Rightarrow 10 \frac{di_L}{dt} + v_c = 3i_1$$

-(1)

KCL at node A :-

$$i + i_1 + i_L = 0 \Rightarrow i_1 = -(i + i_L)$$

KVL in loop (1) :-

$$-i + i_1 = 0 \Rightarrow i = i_1 = -i - i_L$$

$$\Rightarrow 2i = -i_L \Rightarrow i = -0.5 i_L = i_1$$

Substituting in eq (1) :-

$$10 \frac{di_L}{dt} + v_c = \frac{3}{2} i_L$$

Also, capacitor current = $i_c = C \frac{dv_c}{dt} = \frac{dv_c}{dt} \quad [C=1\text{F}]$

$$\therefore 10 \frac{d}{dt} \left[\frac{dv_c}{dt} \right] + \frac{3}{2} \frac{dv_c}{dt} + v_c = 0$$

$$\Rightarrow \frac{d^2 v_c}{dt^2} + 0.15 \frac{dv_c}{dt} + 0.1 v_c = 0$$

Solving the above eqⁿ :-

$$s^2 + 0.15s - 0.1 = 0$$

$$\Rightarrow s = -0.075 \pm j0.3072$$

$$\Rightarrow -\alpha = -0.075 \quad ; \quad \omega_d = 0.3072 \quad [s = -\alpha \pm j\omega_d]$$

Since the roots are complex, the system is underdamped.

$$\therefore v_c(t) = B_1 e^{-\alpha t} \cos(\omega_d t) + B_2 e^{-\alpha t} \sin(\omega_d t)$$

$$v_c(0) = -4.5 \Rightarrow B_1 \cos 0 + B_2 \sin 0 = -4.5$$

$$\Rightarrow B_1 = -4.5$$

Differentiating $v_c(t)$ w.r.t. t ,

$$\frac{dv_c(t)}{dt} = -\alpha B_1 e^{-\alpha t} \cos(\omega_d t) - \omega_d B_1 e^{-\alpha t} \sin(\omega_d t) - \alpha B_2 e^{-\alpha t} \sin(\omega_d t) + \omega_d B_2 e^{-\alpha t} \cos(\omega_d t)$$

$$\Rightarrow \left. \frac{dv_c(t)}{dt} \right|_{t=0} = -\alpha B_1 \cos(0) - \omega_d B_1 \sin(0) - \alpha B_2 \sin(0) + \omega_d B_2 \cos(0)$$

$$\Rightarrow i_c(0)/C = -\alpha B_1 + \omega_d B_2 \Rightarrow \alpha B_1 = \omega_d B_2$$

$$\Rightarrow B_2 = \frac{(-4.5) \times (0.075)}{0.3072} = -1.098$$

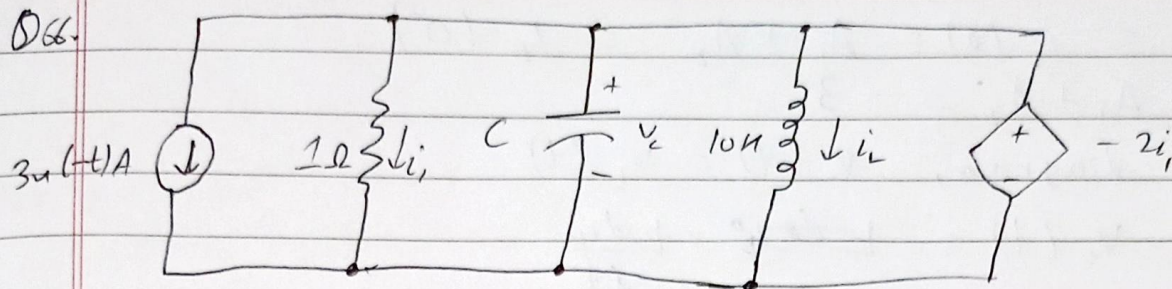
$$\therefore v_c(t) = -4.5 e^{-0.075t} \cos(0.3072t) - 1.098 e^{-0.075t} \sin(0.3072t)$$

a) $v_c(-1) = v_c(0^-) = -4.5 \text{ V}$ (steady-state for both)

b) $v_c(0^+) = v_c(0^-) = -4.5 \text{ V}$ (no sudden voltage changes)

c) $v_c(20) = -4.5 e^{-0.075(20)} \cos(0.3072 \times 20) - 1.098 e^{-0.075(20)} \sin(0.3072 \times 20)$
 $= -1.0243 \text{ V}$

Q6.



a) For overdamped response, $\alpha > \omega_0$.

$$\Rightarrow \frac{1}{2RC} > \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \frac{1}{4R^2C^2} > \frac{1}{LC} \Rightarrow \frac{1}{C} > \frac{4R^2}{L} \Rightarrow C < \frac{L}{4R^2}$$

$$\Rightarrow C < \frac{10}{4(1)^2} = \frac{10}{4} = 2.5$$

$$\Rightarrow \boxed{C < 2.5 \text{ F}}$$

b) For $C = 1 \text{ F}$, the circuit is overdamped.

$$\Rightarrow i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$; s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\text{Where, } \alpha = \frac{1}{2RC} = \frac{1}{2 \times 1 \times 1} = 0.5 \text{ s}^{-1}$$

$$\text{and, } \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10 \times 1}} = 0.316 \text{ s}^{-1}$$

$$\Rightarrow s_1 = -0.5 + \sqrt{0.5^2 - 0.316^2} = -0.1125$$

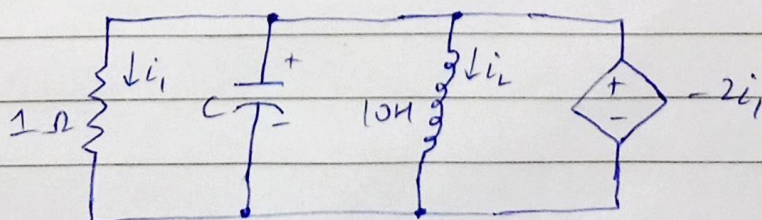
$$s_2 = -0.5 - \sqrt{0.5^2 - 0.316^2} = -0.887$$

$$\therefore i_L(t) = A_1 e^{-0.1125t} + A_2 e^{-0.887t} \quad \text{--- (1)}$$

At $t = 0^-$, inductor is short-circuited and capacitor is open-circuited.

$$\Rightarrow i_L(0^-) = -3 \text{ A} ; v_C(0^-) = 0 \text{ V}$$

For $t > 0$:



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$$\text{Eq (1)} :- i_L(0) = A_1 + A_2 = i_L(0^-)$$

$$\Rightarrow A_1 + A_2 = -3$$

From diagram, $v_c(t) = v_L(t)$

$$\Rightarrow v_c(t) = L \frac{di_L}{dt}$$

$$v_c(0) = v_c(0^-) = 0V$$

$$\Rightarrow L \left. \frac{di_L}{dt} \right|_{t=0} = 0 \Rightarrow \left. \frac{di_L}{dt} \right|_{t=0} = 0$$

Differentiating eq (1) :-

$$\left. \frac{di_L}{dt} \right|_{t=0} = -0.1125A_1 e^{-0.1125t(0)} - 0.887A_2 e^{-0.887t(0)} = 0$$

$$\Rightarrow 0.1125A_1 + 0.887A_2 = 0$$

Also, $A_1 + A_2 = -3$

$$\therefore A_1 = -3.4358 \quad ; \quad A_2 = 0.4358$$

$$\therefore \boxed{i_L(t) = -3.4358 e^{-0.1125t} + 0.4358 e^{-0.887t} \text{ A}}$$